

AIR (Advice & Intelligence Relationship) System Architecture Design – V3.0

Version: 3.0

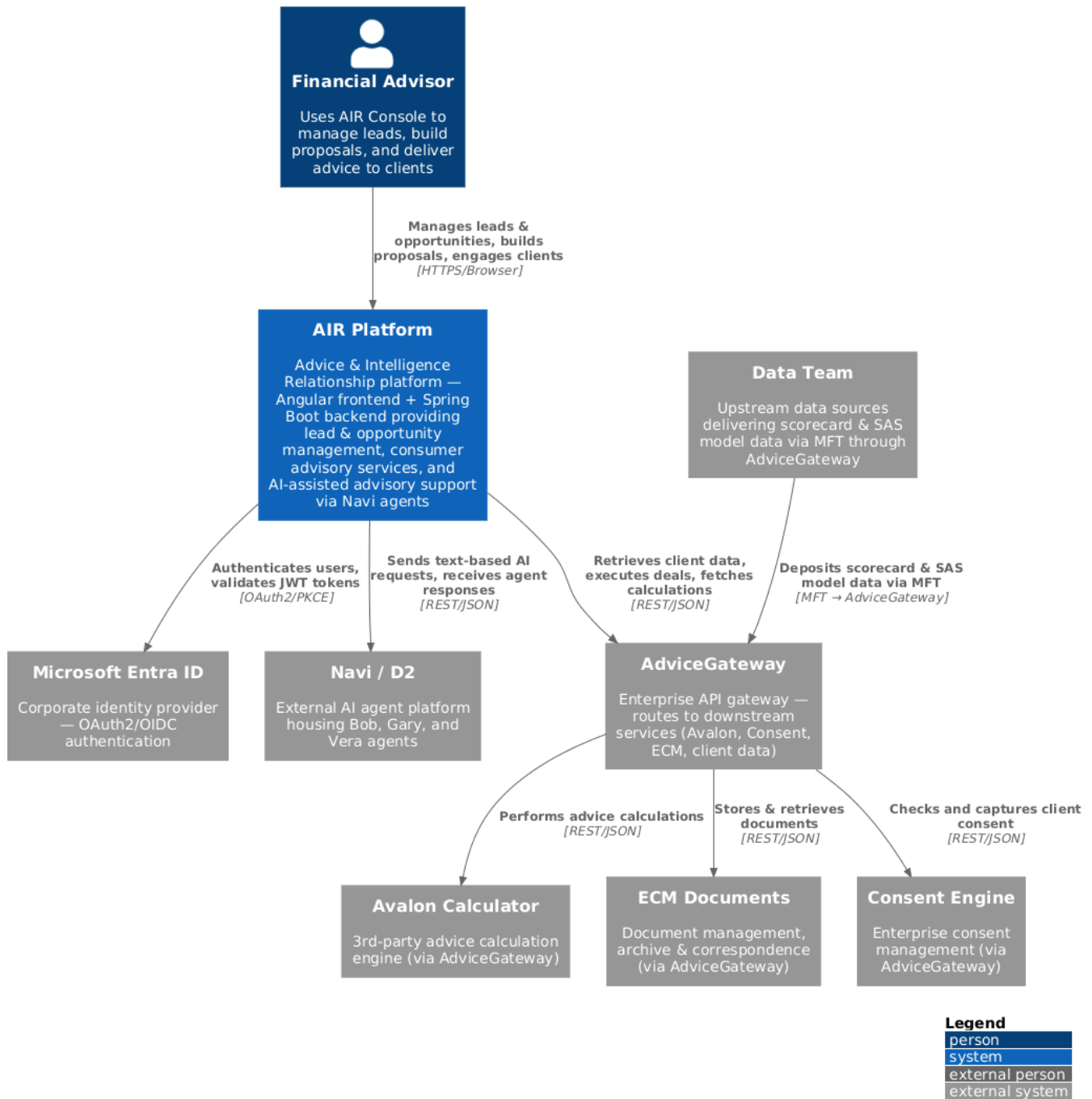
Based on: System Architecture Design V2.3 + Architecture Simplification
(callback-based, Navi/D2 agent delegation)

Date: June 2026

Level 1: System Context Diagram

Shows the AIR system in its environment, with external actors and systems.

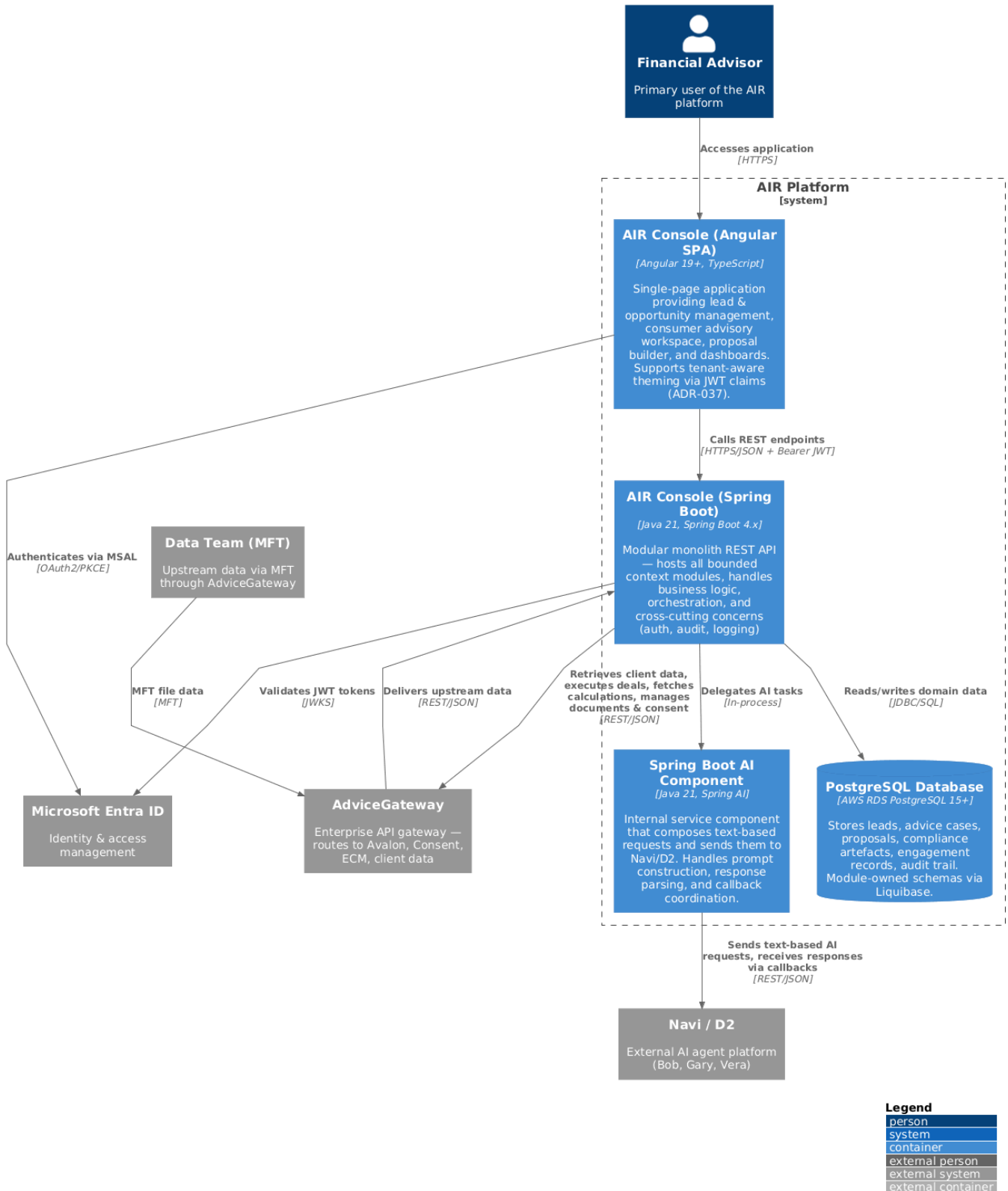
AIR System Context Diagram



Level 2: Container Diagram

Shows the major deployable units and their interactions.

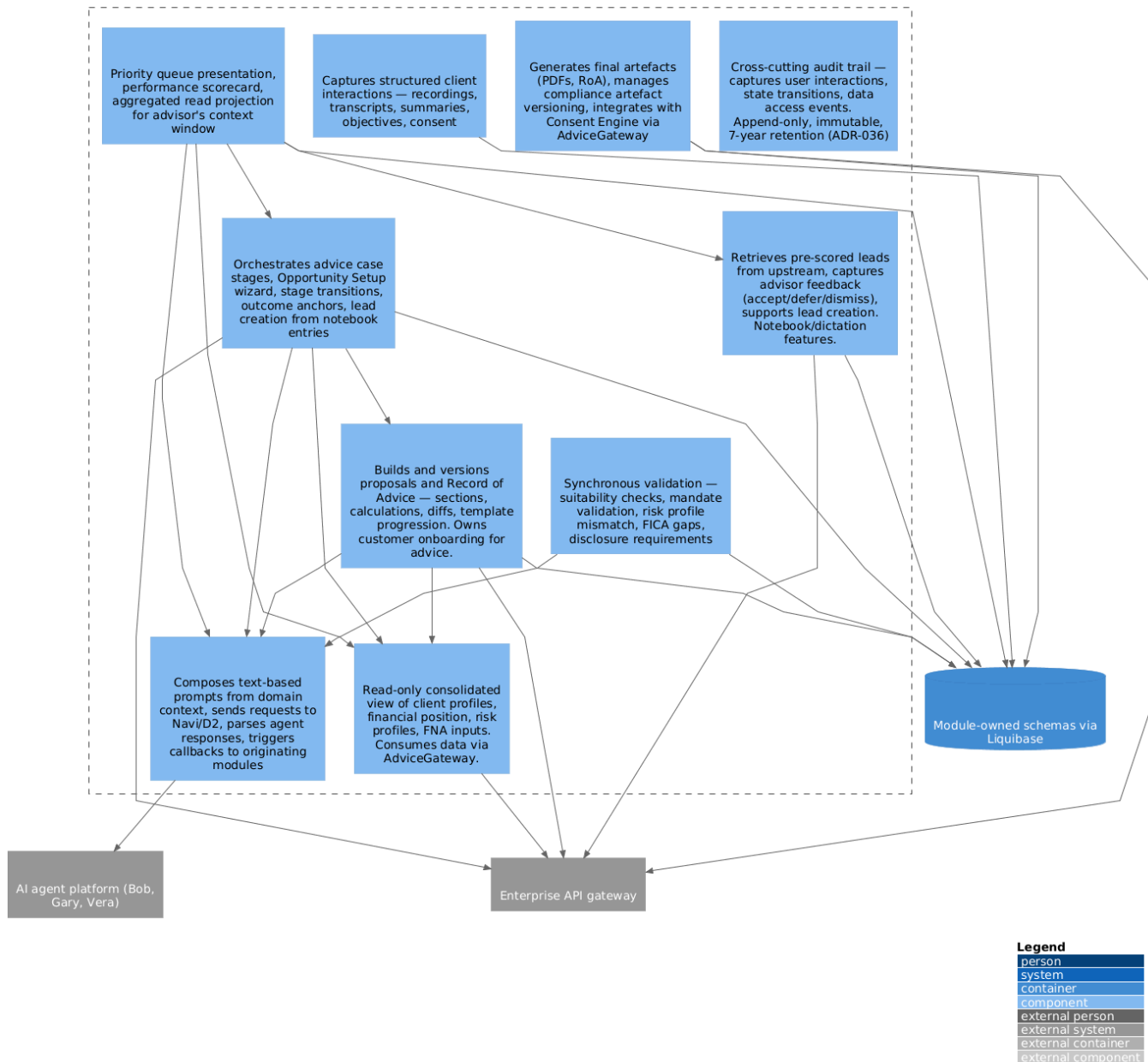
AIR Container Diagram



Level 3: Component Diagram – AIR Console Backend (Modular Monolith)

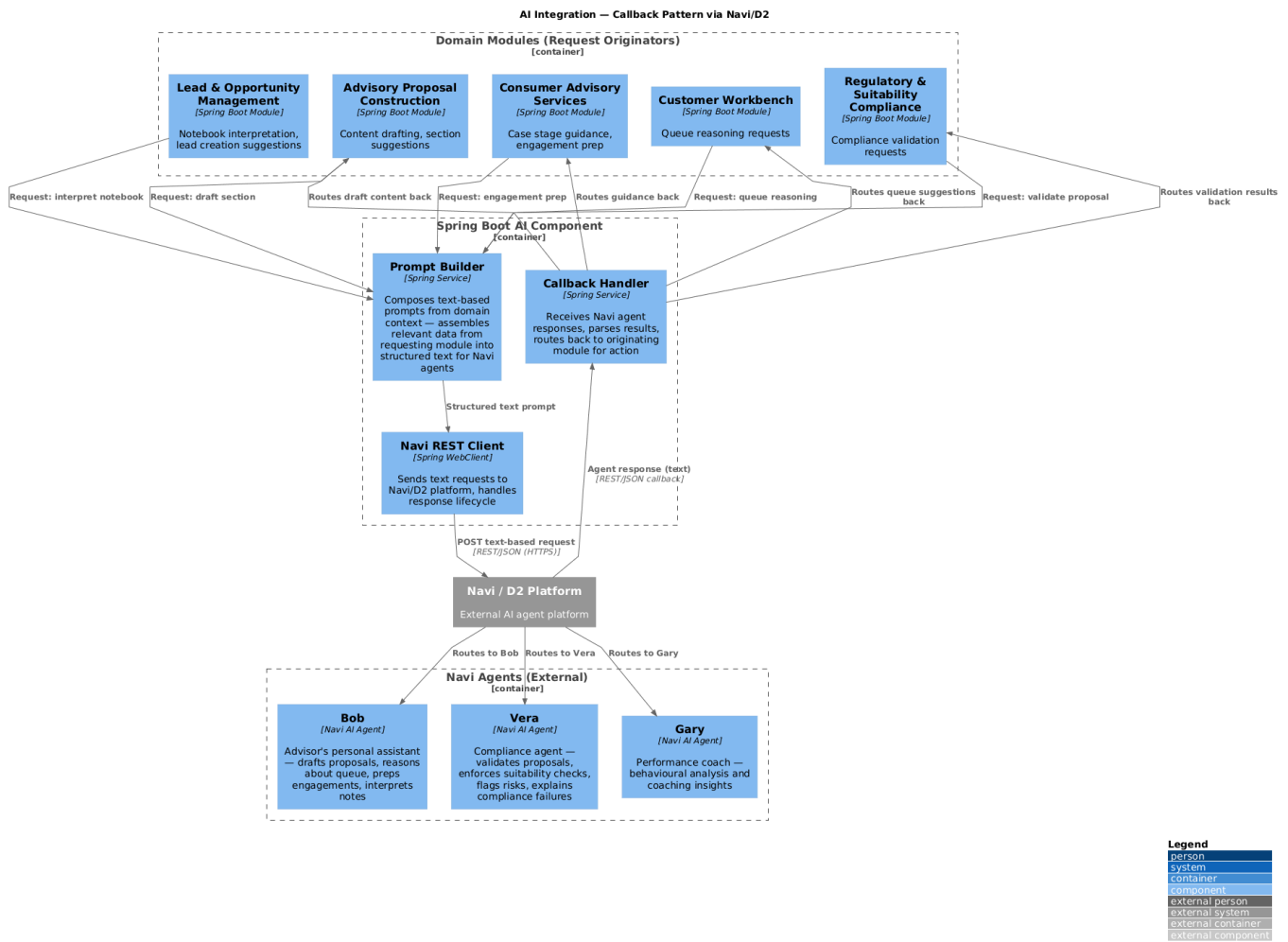
Shows the internal module structure of the monolith and its bounded contexts.

AIR Console Backend — Modular Monolith Component Diagram



Level 3: Component Diagram – AI Integration (Callback Pattern)

Focused view of how the AIR backend interacts with Navi/D2 agents via the Spring AI component.



Level 4: Deployment Diagram

Shows the physical deployment topology on AWS.

CI/CD Pipeline Flow

Illustrates the trunk-based development and deployment pipeline with governance gates.

Cross-Cutting Concerns

Application Interaction Audit Trail (ADR-036)

A cross-cutting audit service captures all significant user interactions across the AIR platform. Implemented via Spring AOP interceptors on controller/service boundaries.

Audit Event Schema:

Field	Description
eventId	UUID
timestamp	ISO-8601 (UTC)
actorId	Entra ID user principal
actorRole	Role from JWT claims
action	Canonical action (e.g. VIEW_CLIENT_PROFILE , APPROVE_CASE)
module	Originating bounded context
resourceType	Entity type (e.g. AdviceCase , Proposal)
resourceId	Entity identifier
outcome	SUCCESS , DENIED , ERROR
metadata	JSON context (state changes, sections edited)
correlationId	Links related audit entries
clientId	Client identifier for per-client queries

Key characteristics: Append-only, immutable, 7-year retention (FAIS compliance). Hot storage 12 months in PostgreSQL audit schema, cold storage in S3 (Parquet).

Tenant-Aware UI Theming (ADR-037)

The Angular SPA supports runtime brand skinning based on authenticated user's tenant/brand affiliation (resolved from JWT tenant_brand claim). Single build artefact serves all brands via CSS custom properties over Angular Material's M3 design token system.

Supported Tenants (MVP):

- fnb-internal – FNB-branded advisors (default)
 - Future: ifa-partner-* – external IFA partners (white-label)
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Key Architecture Characteristics

- **Simplified Container Model:** Angular SPA → Spring Boot monolith → PostgreSQL, with AdviceGateway as the single external integration point and Navi/D2 as the AI platform
- **Modular Monolith:** All bounded contexts deployed as a single unit with explicit module boundaries for future decomposition
- **8 Active Bounded Contexts:** Customer Workbench, Lead & Opportunity Management, Customer Relationship & Insights, Consumer Advisory Services, Advisory Proposal Construction, Regulatory & Suitability Compliance, Session Dialogue & Contact, Document Services
- **No Event Bus:** Cross-module communication uses direct method calls within the monolith. AI agent interactions use a **callback pattern** – the Spring AI component sends text requests to Navi/D2 and receives responses via REST callbacks
- **Navi/D2 as External AI Platform:** All AI agent capabilities (Bob, Gary, Vera) are hosted externally on the Navi/D2 platform. The AIR backend sends text-based requests via the Spring Boot AI component and receives structured responses
- **Bob (via Navi):** Advisor's personal assistant – drafts proposals, reasons about queue priority, preps engagements, interprets notebook entries
- **Vera (via Navi):** Compliance agent – validates proposals, enforces suitability checks, flags risks, explains compliance failures
- **Gary (via Navi):** Performance coach – behavioural analysis and coaching insights (deferred to future phase)
- **AdviceGateway as Integration Hub:** All external service calls (Avalon, ECM, Consent Engine, client data) route through AdviceGateway – single point for circuit breaking, retry, and observability
- **Data Team / MFT Integration:** Upstream data (scorecards, SAS models) flows from Data Team sources through MFT into AdviceGateway and onward to the AIR backend
- **Callback-Based AI Interaction:** Domain modules request AI assistance → Spring AI component builds text prompt → sends to Navi/D2 → Navi routes to appropriate agent (Bob/Vera/Gary) → agent response returned via callback → routed back to originating module
- **AI Cannot Change State:** Agents suggest, flag, and advise – humans approve and transition
- **Application Interaction Audit Trail:** Cross-cutting service capturing all user interactions via Spring AOP. Append-only, immutable, 7-year retention (ADR-036)
- **Tenant-Aware UI Theming:** Runtime brand skinning from JWT claims. Single codebase serves multiple brands (ADR-037)
- **Feature Toggles:** Decouple code deployment from business release
- **PR Environments:** Ephemeral namespaces with mock data for rapid feedback
- **Module-Owned Data:** Each module manages its own schema via Liquibase
- **CI/CD Governance:** Acceptance criteria sign-off before development. Test

reports at PR and release stages. Daily 5am production deployments.

Diagram Summary

Level	Diagram	Purpose
C4 Level 1	System Context	Shows AIR in its ecosystem – advisor, AdviceGateway (routing to Avalon/ECM/Consent), Entra ID, Navi/D2, Data Team
C4 Level 2	Container	Shows deployable units – Angular SPA, Spring Boot monolith (with Spring AI component), PostgreSQL
C4 Level 3	Component (Backend)	Shows internal bounded context modules and their relationships
C4 Level 3	Component (AI Integration)	Shows callback pattern – prompt building, Navi REST client, callback handler, agent routing
C4 Level 4	Deployment	Shows AWS topology – EKS pods, RDS, CloudFront, S3
Supplementary	CI/CD Pipeline	Shows trunk-based development with governance gates and daily deployment
Supplementary	Cross-Cutting Concerns	Documents audit trail and tenant-aware theming

Glossary

Term	Definition
AIR	Advice & Intelligence Relationship – the platform being designed
AdviceGateway	Enterprise API gateway through which all external service calls are routed (Avalon, ECM, Consent, client data)
Navi / D2	External AI agent platform housing Bob, Gary, and Vera agents
Bob	AI personal assistant agent (hosted on Navi) – advisor's co-pilot for proposal drafting, queue reasoning, engagement prep, notebook interpretation
Vera	AI compliance agent (hosted on Navi) – proposal validation, suitability enforcement, compliance failure explanations

Term	Definition
Gary	AI performance coach agent (hosted on Navi) – behavioural analysis and coaching insights (deferred)
Avalon	3rd-party advice calculation engine – accessed via AdviceGateway
ECM	Enterprise Content Management – document storage, archive & correspondence (accessed via AdviceGateway)
Consent Engine	Enterprise consent management – checks/captures client consent (accessed via AdviceGateway)
MFT	Managed File Transfer – delivers batch data (scorecard, SAS models) from Data Team through AdviceGateway
Callback Pattern	AI interaction model – AIR sends request to Navi, Navi processes via appropriate agent, returns response via REST callback to AIR
Spring AI Component	Internal service that composes text prompts, communicates with Navi/D2, and routes responses back to domain modules
Entra ID	Microsoft Entra ID – corporate identity provider (OAuth2/OIDC)
RoA	Record of Advice – regulatory documentation of advice given
FNA	Financial Needs Analysis – client assessment inputs
FAIS	Financial Advisory and Intermediary Services Act – SA regulatory framework requiring 7-year audit retention
IFA	Independent Financial Advisor – external (non-FNB) advisor using AIR under white-label branding
HPA	Horizontal Pod Autoscaler – Kubernetes auto-scaling mechanism

Document version: 3.0 – Simplified Architecture (Callback-based, Navi/D2 delegation)

Based on: V2.3 + Architecture simplification decisions

Date: June 2026